Set the kth bit.

5 4 3 2 1 0

n = 1 1 0 0 1 0 , k = 2

O/P = 1 1 0 1 1 0

0 | 1 = 1

1 | 1 = 1

2^4 16 + 32 = 50

n

1 1 0 0 1 0

OR 0 0 0 1 0 0 — OR mask 100

1 1 0 1 1 0

1000 mask = (1 << k)

res = n | (1 << k)

1

10

100

n = 1 1 0 1 1 0 k = 2

1

1 1 0 0 1 0

mask 1 1 1 0 1 1

n = 1 1 0 1 1 0

mask = 1 1 1 0 1 1

1 1 0 0 1 0

... 0 0 1 0 0

... 1 1 1 0 1 1 mask = ~(1<<k)

res = n & (~(1<<k)) O/P 0

1 ^ 1 0

0 ^ 1 1

n = 1 1 0 1 1 0, k = 2

XOR (0 0 0 1 0 0)

1 1 0 0 1 0

res = n ^ (1<< k)

n = 1 0 1 1 (11) n = 1 0 0 (8)

O/P = 3 O/P = 1

i: 0 → 31

if (n | (1<<i)) != n — cnt++ — O(32)

n = 1 0 1 1 → 1 0 1 → 1 0 → 1 → 0

while (n > 0)

if n%2 == 1

cnt++

n = n>>1

O(pos. of the leftmost set bit)

n = 1 0 1 0 0 (20) 16+2+1

n-1 = 1 0 0 1 1 1 0 1 0 0

AND 1 0 0 0 0 1 0 0 1 1

8+4+2+1

n = 1 0 0 0 0

n-1 = 0 1 1 1 1

AND =

0 0 0 0 0

(n & (n-1)) ⇒ is the same as n with rightmost set bit becoming 0.

while n!=0

n=(n & (n-1))

T.C = O(no. of set bits)

1

10

100

1000

10000

100000

$2^0 = 1$

n=1 O/P: True

(n & (n-1) == 0)